

Predicting Reading Comprehension in Children with Different Types of Reading Disorders

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Abstract

Several models of reading comprehension have been proposed. However, few studies have included profiles of reading ability beyond typical readers and dyslexia.

This project sought to compare models of reading comprehension across several profiles of reading disorder (dyslexia, poor comprehenders, and mixed/unknown reading disorder) and typical readers.

Decoding was modeled as a single and two factor construct; there was no best decoding model for all of the groups. The Simple View of Reading was the best-fit model for the typical reader, dyslexic, and poor comprehender groups; whereas, the Componential Model of Reading was a moderately better model for the mixed/unknown reading disorder group.

The key finding is that a general model of reading comprehension cannot be assumed to fit students with different types of reading disorder.

Introduction

- 15.4% of children with a reading disorder have poor reading comprehension (Catts, Hogan, & Fey, 2003).
- Historically, reading comprehension has been modeled using the Simple View of Reading (Gough & Tunmer 1986) and more recently other models have been presented, including the Componential Model of Reading (Joshi & Aaron, 2000).
- Children with different profiles of reading disorders are rarely included in explorations of models for reading comprehension; specifically children with poor comprehension and children with a mixed or unknown reading disorder profile.

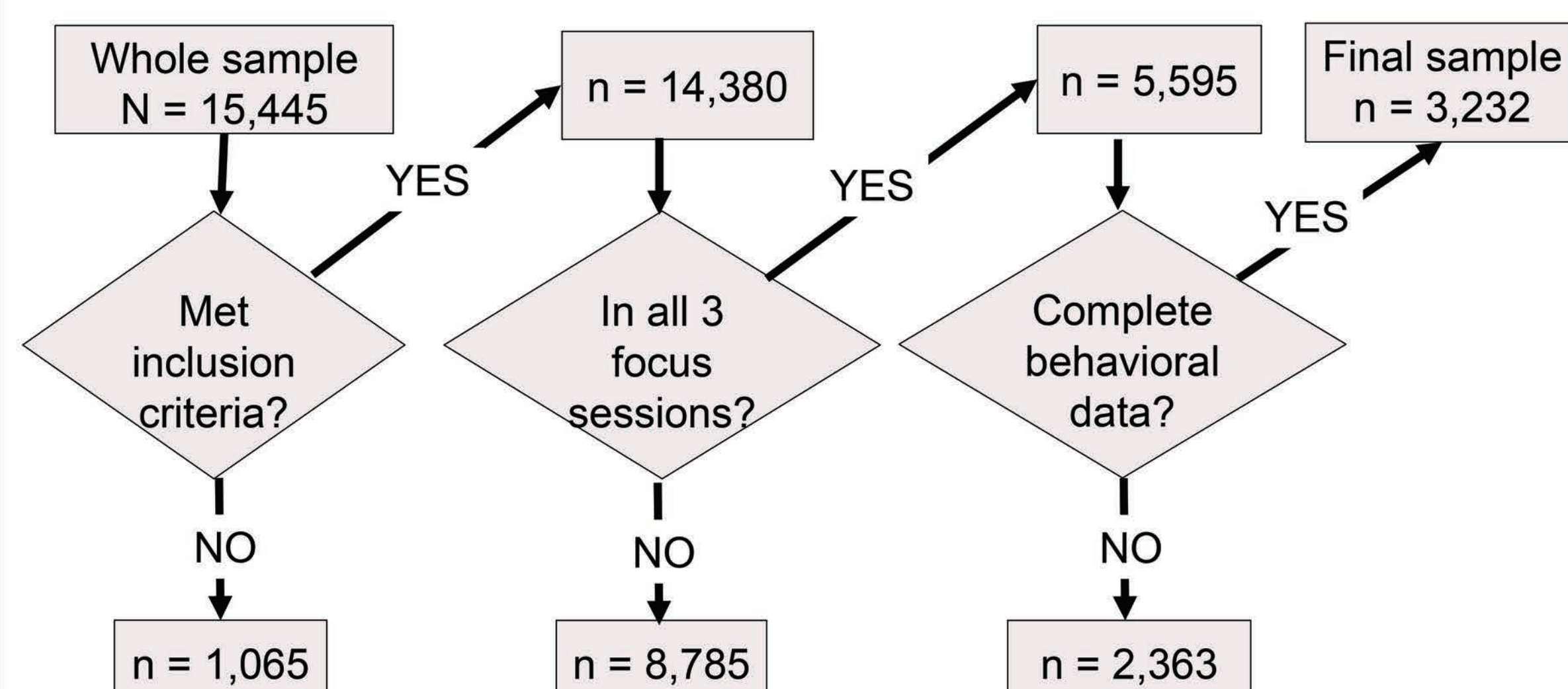
Objective of this study. This project modeled variations of the Simple View of Reading and the Componential Model of Reading to determine which model best fits for children with different types of reading disorders compared to typical readers. We had two research questions:

- Should decoding in the Simple View of Reading be a single or two factor construct?
- Which model, Simple View of Reading or Componential Model of Reading, is a better fit for each reading profile?

Methods

Data source. The Avon Longitudinal Study of Parents And Children (ALSPAC; Boyd et al., 2013) is a prospective birth cohort. Data collection began the early 1990s and is ongoing. The sample size at year 1 was 14,684. Behavioral data collection began at age 7 with focus sessions. A selection of the original participants plus new recruits were invited to attend these sessions.

Participant selection. We selected participants who meet inclusion criteria (normal hearing, no diagnosis of ASD, nonverbal IQ ≥ 72 , and one twin per pair only) and participated in "Focus" sessions at ages 7, 8, and 9. Below is a flow chart of participant selection.



Methods

Reading profile classification. Children were divided into four reading profiles using the following criteria:

- Typical reader (TR) – All reading measures were greater than -1 z-scores
- Dyslexia (DYS) – Participant scored less than -1 z-score on at least three of the following measures: Single word reading age 7, single word reading age 9, phoneme deletion, or nonword reading
- Poor comprehension (PC) – Reading comprehension was less than -1 z-score AND participant did not meet the criteria of DYS
- Mixed/unknown reading disorder (MRD) – Three additional reading measures were less than -1 z-scores AND participant was not classified as DYS or PC

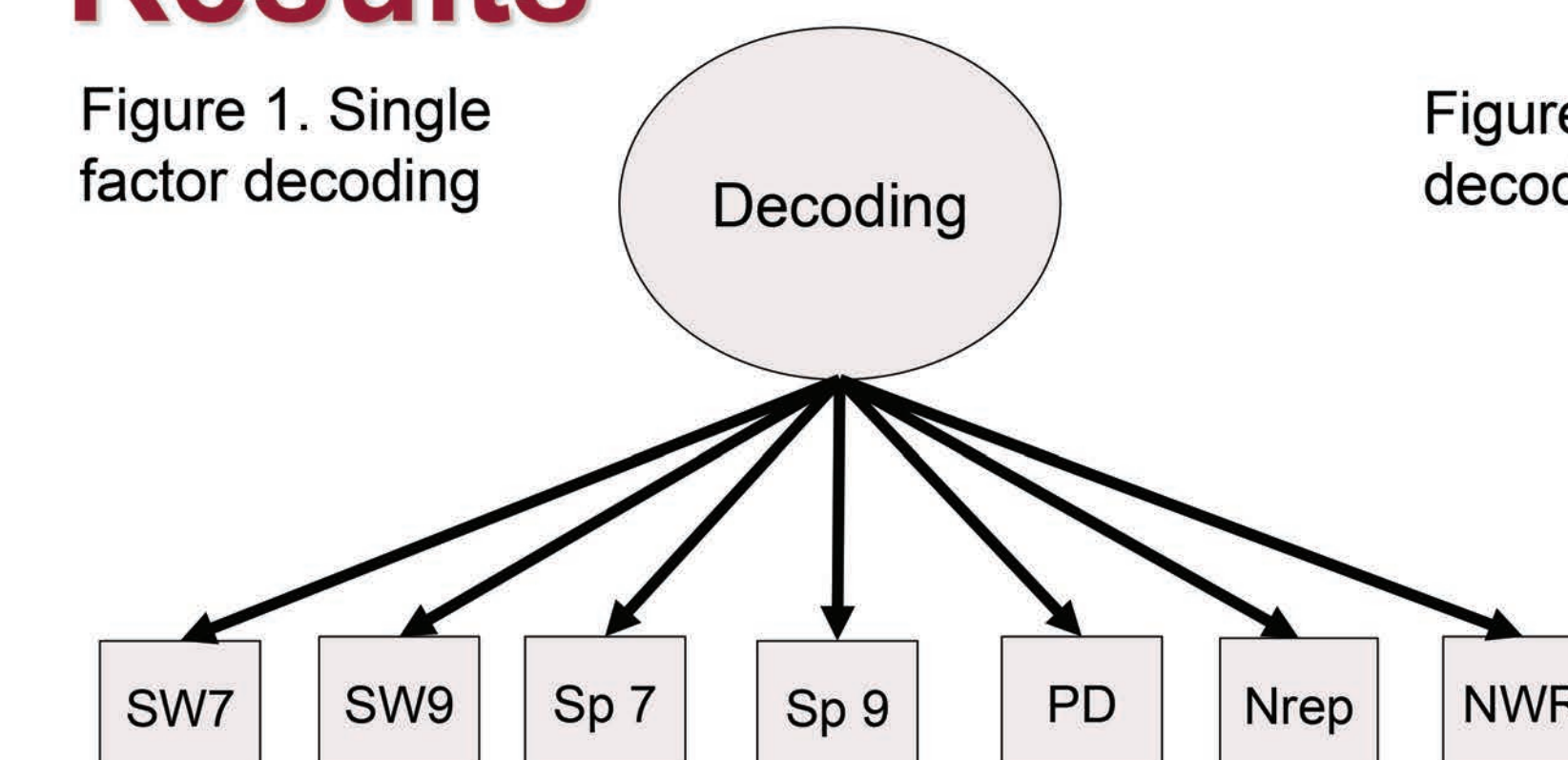
Measures. Descriptive, reading, and other behavioral measures were selected from the ALSPAC and spanned from prenatal to 9 years. Three measures came from the Neale Analysis of Reading Ability (Neale, 1997). Rate and accuracy were only used for group classification. Table 1 has the descriptive statistics for all measures used for classification and in analysis.

Analysis. Confirmatory factor analysis and structural equation modeling were completed in R (2016) using lavaan (Rosseel, 2012) with maximum likelihood estimation with robust (Huber-White) statistics. Prior to fitting models, we tested for measurement invariance to determine if there were equivalent constraints across groups. We used the following goodness-of-fit statistics and criteria to judge model fit:

- Tucker Lewis Index (TLI) – A value of 0.80 is good fit, and 0.90 is great fit.
- Root Mean Square Error Approximation (RMSEA) – A value less than 0.08 is evidence of decent fit.
- Standardized Root Mean Square Residual (SRMR) – A value less than 0.08 is considered good fit.

Results

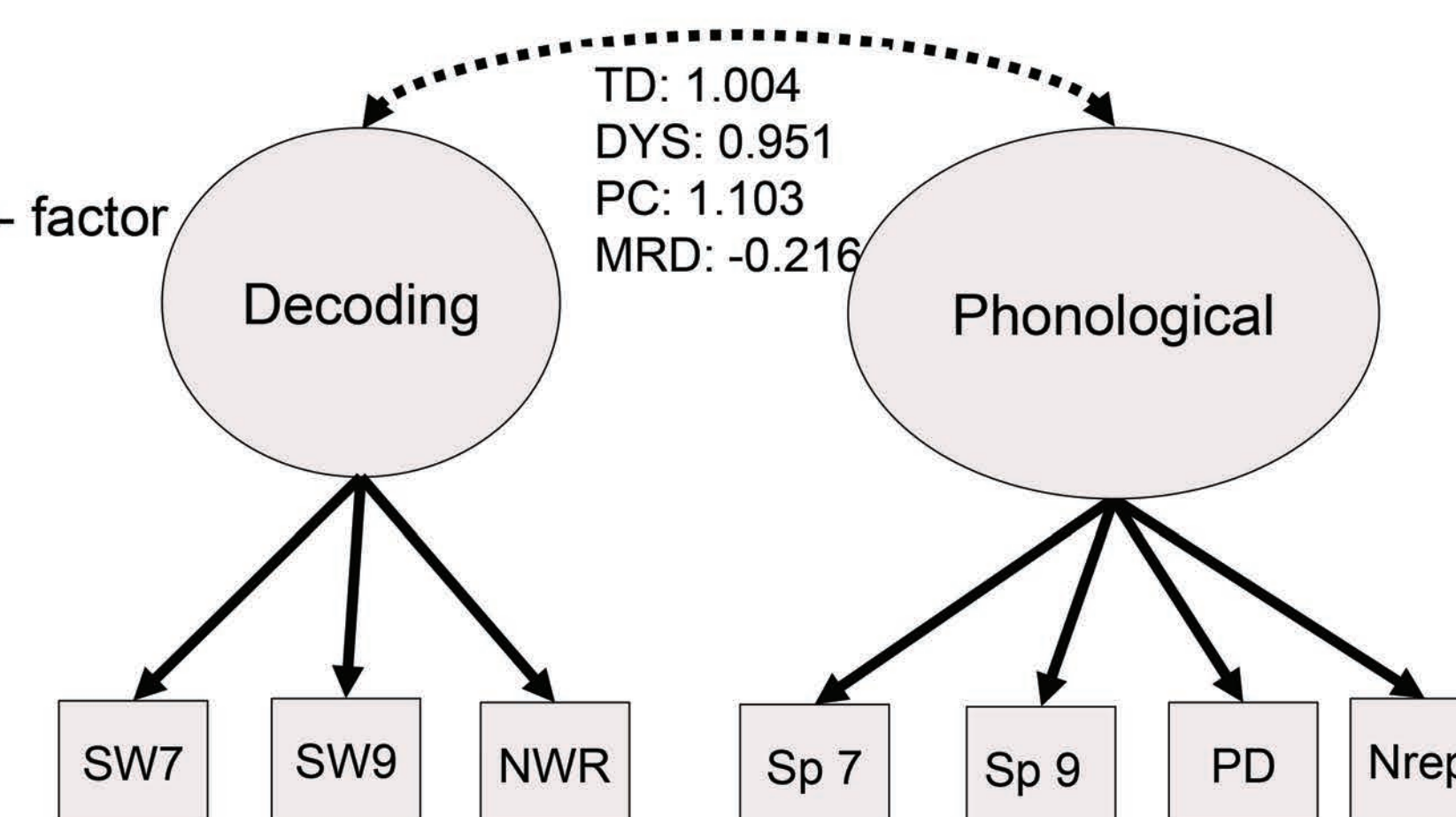
Figure 1. Single factor decoding



Goodness of fit statistic	TD	DYS	PC	MRD
TLI	0.917	0.742	0.767	-0.689
RMSEA	0.097	0.093	0.145	0.165
SRMR	0.036	0.066	0.066	0.077

- Nonword repetition was not a useful variable for TD or DYS groups.
- Phonemic decoding was nonsignificant for DYS.
- For the MRD group, a single factor was inappropriate.

Figure 2. Two-factor decoding



Goodness of fit statistic	TD	DYS	PC	MRD
TLI	0.909	0.711	0.775	0.760
RMSEA	0.101	0.098	0.143	0.062
SRMR	0.036	0.066	0.065	0.058

- Model comparisons to the single factor indicated that a 2-factor was not a better model for TD, DYS, PC
- It was a better model for MRD; although only nonword repetition and spelling at 9 were contributing paths for the MRD group.

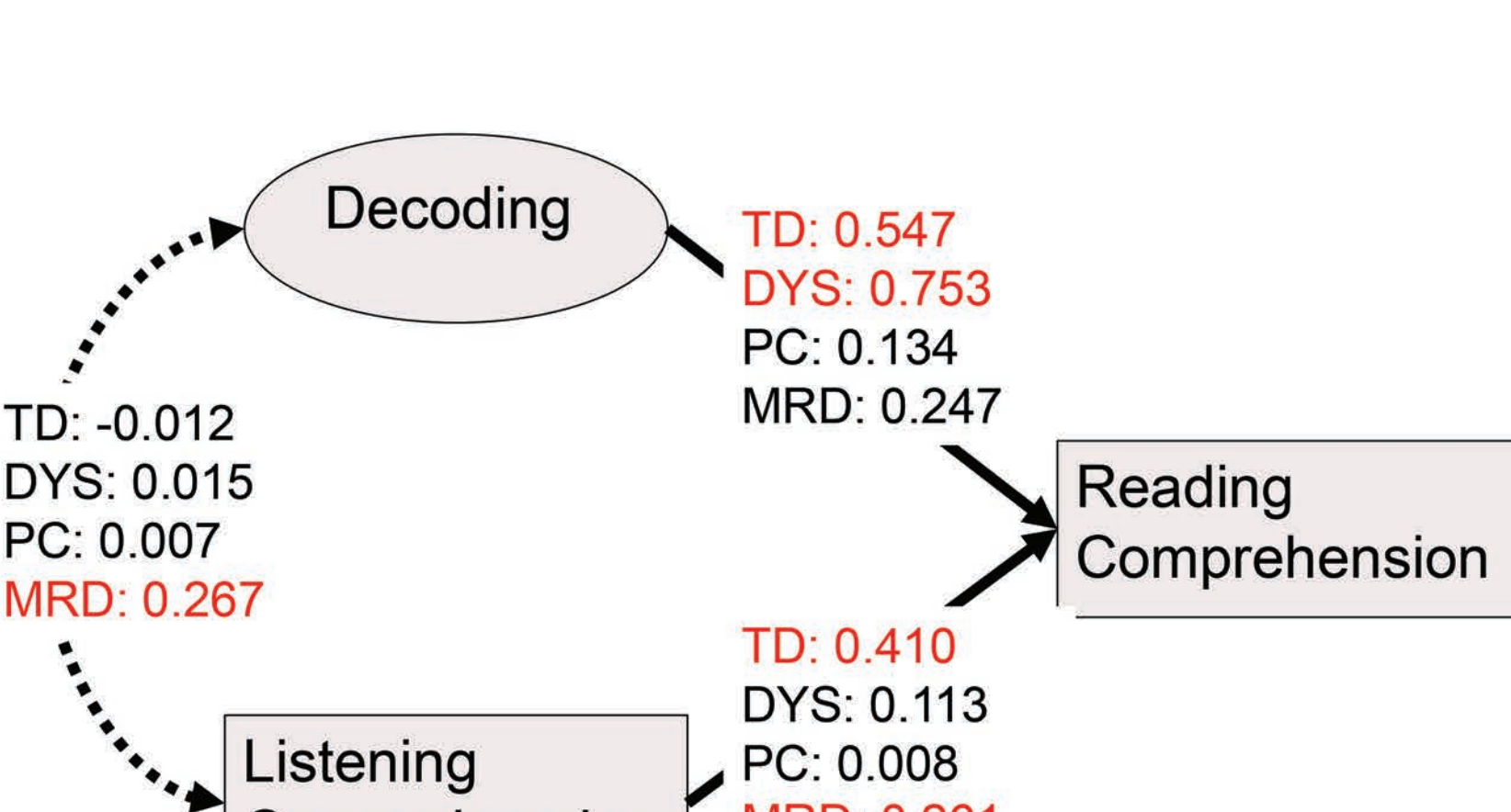


Figure 3. Simple View of Reading

Goodness of fit statistic	TD	DYS	PC	MRD
TLI	0.913	0.788	0.740	0.237
RMSEA	0.085	0.085	0.116	0.111
SRMR	0.038	0.064	0.065	0.082

- This model explained between 36% (MRD) and 43% (PC) total variance.
- We ran the model with a single factor and 2-factor Decoding for the MRD group, but the 2-factor model did not converge after 1000 iterations.

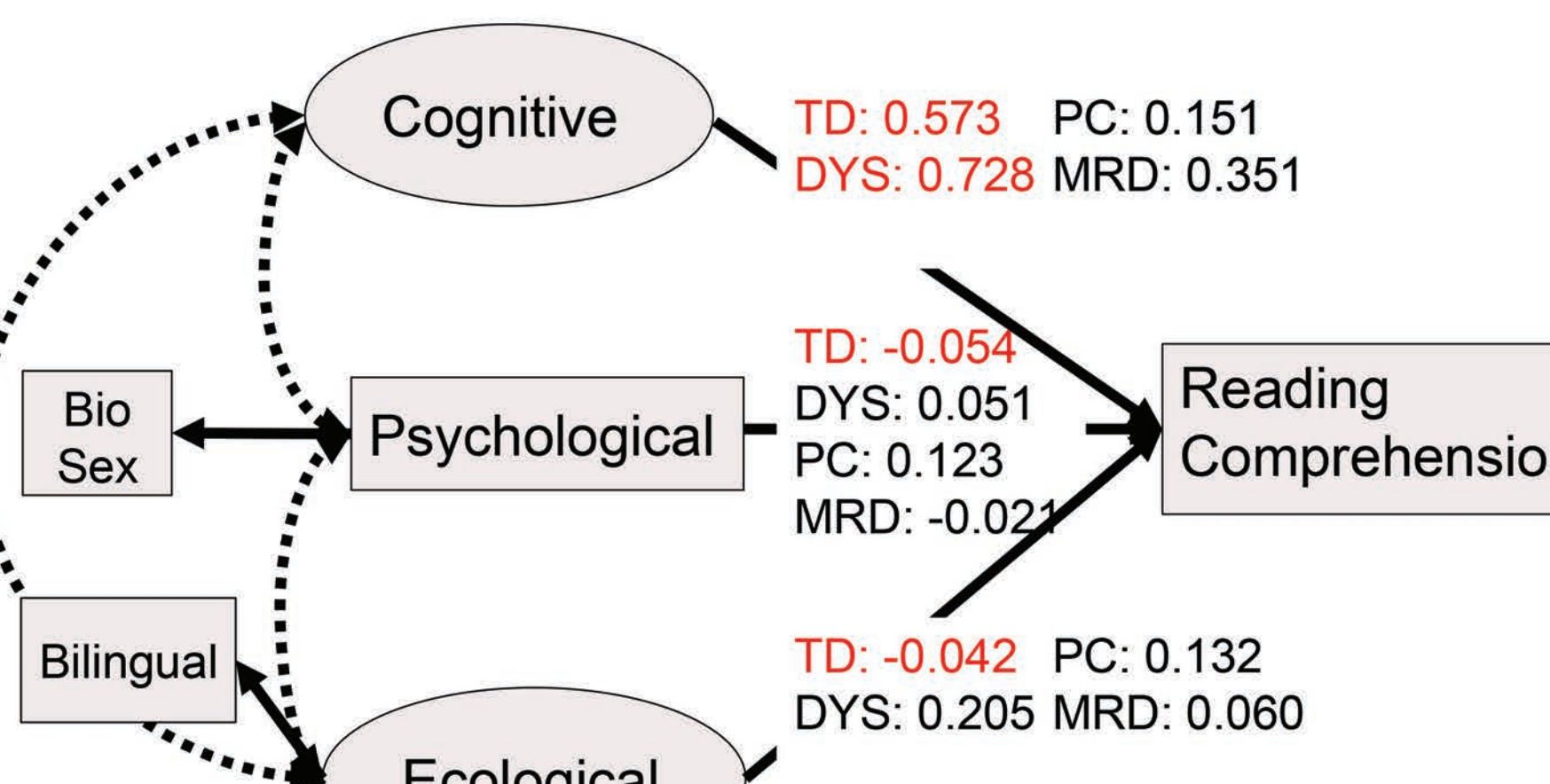


Figure 4. Componential Model of Reading

Goodness of fit statistic	TD	DYS	PC	MRD
TLI	0.789	0.774	0.623	0.318
RMSEA	0.088	0.065	0.103	0.083
SRMR	0.060	0.060	0.074	0.078

- We added attention and vocabulary to the cognitive component.
- Attention did not add to the model.
- Explain less than the SVR model for the PC and MRD groups.

Methods

Table 1.

Descriptive statistics for demographic, reading, and behavioral measures by reading profile.

Variable	Typical Reader	Dyslexia	Poor Comprehender	Mixed/Unknown Reading Disorder
Demographic				
N	2,582	251	208	191
Biological sex ^a	44.00%	57.00%	58.00%	54.00%
Bilingual language status ^b	2.40%	1.60%	0.50%	2.10%
Nonverbal IQ	104.76 (15.89)	94.99 (13.91)	92.99 (14.51)	99.44 (15.33)
Maternal education ^c	4.00	3.00	3.00	3.00
Reading				
Single word age 7	32.20 (7.05)	14.99 (3.67)	21.76 (5.94)	20.28 (4.52)
Spelling age 7	30.50 (10.36)	11.84 (8.44)	5.94 (18.84)	4.52 (13.51)
Single word reading age 9	8.59 (1.35)	3.10 (1.76)	6.14 (2.09)	5.80 (1.73)
Nonword reading age 9	6.07 (2.03)	1.48 (1.14)	4.23 (1.90)	3.15 (1.67)
Spelling age 9	11.65 (2.13)	4.90 (2.89)	7.91 (3.03)	7.16 (2.59)
Rate (NARA)	109.05 (10.05)	09.80 (11.60)	97.18 (10.75)	95.43 (9.01)
Accuracy (NARA)	109.87 (10.15)	84.06 (8.13)	91.57 (7.83)	94.17 (5.46)
Reading comprehension (NARA) ^d	105.58 (8.81)	84.83 (7.57)	85.25 (3.29)	95.95 (5.84)
Phonological awareness				
Phoneme deletion	23.37 (8.22)	9.03 (5.79)	15.92 (7.09)	14.60 (6.60)
Nonword repetition	7.87 (2.26)	5.57 (2.39)	6 (2.39)	6.23 (2.23)
Language				
Receptive language	7.79 (1.94)	6.91 (1.67)	6.39 (1.66)	7.52 (1.74)
Vocabulary	12.16 (4.21)	8.69 (3.17)	8.27 (3.11)	10.28 (3.52)
Attention				
Sustained attention - Sky Search	4.21 (4.98)	5.51 (1.77)	5.54 (1.89)	5.22 (1.51)

Notes. ^a Percent male. ^b Percent bilingual. ^c Median. ^d Outcome variable. Mean (SD). NARA – Neale Analysis of Reading Ability (Neale, 1997).

Conclusion

This preliminary work has several implications for modeling reading comprehension in children with a wide range of reading skills.

- Firstly, we cannot assume that the structure of constructs such as decoding are the same across groups.
 - Children with dyslexia, poor comprehension, and mixed/unknown reading disorder have a different factor structures for decoding compared to typical readers.
- Secondly, these results suggest that the Simple View of Reading was a better fitting model of reading comprehension for the typical readers, dyslexic, and poor comprehender groups.
 - Although the Componential Model of Reading was a moderately better model for the mixed/unknown reading disorder group.

For all measurement invariance tests, we found that the groups were not equivalent for factor structure, intercepts, loadings, and means. This means that each group was better modeled separately than combined with another group.

This study is limited by the number of indicators available for the certain components of the models (e.g., only one variable for language comprehension), which does limit the conclusions we can draw from these findings.

Take together our findings highlight the variability in reading comprehension across different profiles of reading disorder and the need to investigate reading comprehension within each profile, so as to give us a better and more accurate model for each profile.

Acknowledgements

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For more information, visit:

<https://tinyurl.com/lancaster-F32>